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# Lab 2

```
import numpy as np
import matplotlib.pyplot as plt
```

```
# Parameters
N = 20         # Number of bits
sps = 10       # Samples per symbol
fc = 2         # Carrier frequency (Hz)
fs = sps * 10  #Sampling frequency
t = np.arange(0, N, 1/sps) # time vector
```

```
# 1. Source bits
bits = np.random.randint(0,2,N)
symbols = 2*bits - 1  #BPSK mapping:0→-1, 1→+1
baseband = np.repeat(symbols, sps)
```

```
# 2. Carrier
carrier = np.cos(2*np.pi*fc*t)
```

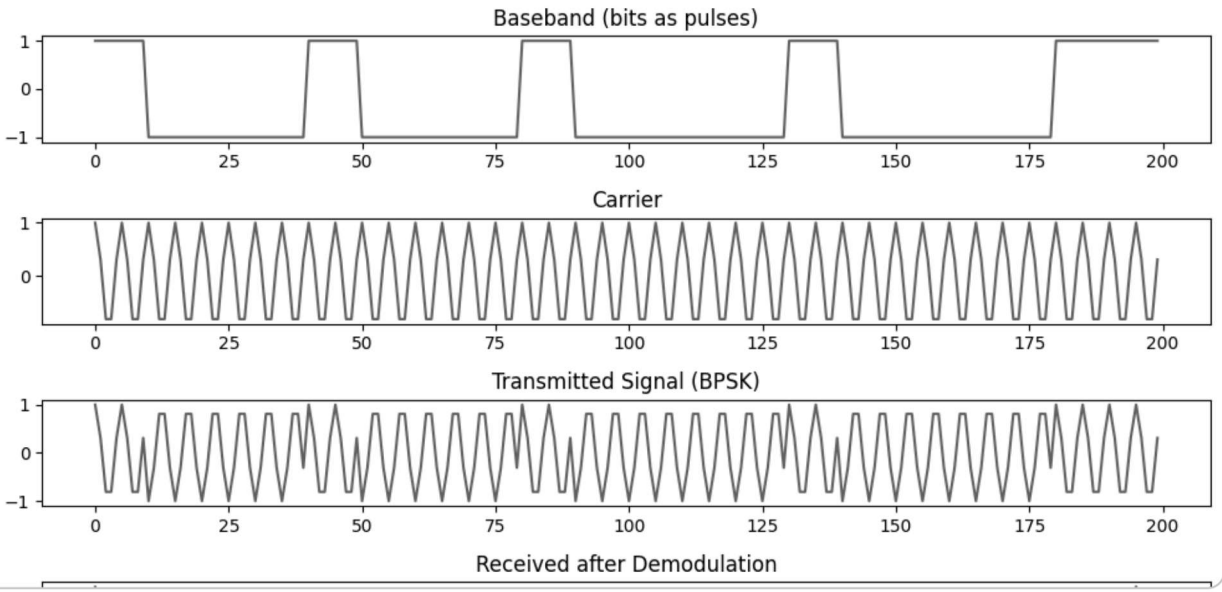
```
# 3. Modulated signal
tx = baseband * carrier
```

```
# 4. Add simple noise
noise = 0.3 * np.random.randn(len(tx))
rx = tx + noise
```

```
# 5. Demodulate (multiply by carrier again)
demod = rx * carrier
```

```
#  Plots
plt.figure(figsize=(10,6))
plt.subplot(4,1,1); plt.plot(baseband); plt.title("Baseband (bits as
plt.subplot(4,1,2); plt.plot(carrier); plt.title("Carrier")
plt.subplot(4,1,3); plt.plot(tx); plt.title("Transmitted Signal (BPSK
plt.subplot(4,1,4); plt.plot(demod); plt.title("Received after Demodu
```

```
plt.tight_layout()
plt.show()
```




```
# upload Darw.io image
from IPython.display import Image, display
display(Image(filename='/content/Annette_Bazan_ITAI4
```

